

KAKE, AK, USA

WMO: 703855

Lat: 56.961N

Lon: 133.910W

Elev: 52

StdP: 100.70

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25369

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.1	-8.1	-17.8	0.8	-7.9	-16.2	0.9	-6.5	9.9	3.7	8.8	4.2	2.7	70	0.646

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.2	21.0	14.4	18.7	13.8	17.4	12.8	15.2	19.0	14.5	17.7	13.7	16.5	2.5	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.2	9.5	16.3	12.7	9.2	15.7	12.3	9.0	15.3	42.4	19.3	40.4	17.6	38.3	16.5	19.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
8.4	7.5	6.6														
			DB	-13.7	24.1	2.3	2.5	-15.3	25.9	-16.6	27.4	-17.9	28.8	-19.5	30.7	
			WB	-14.2	16.4	2.1	1.0	-15.7	17.1	-17.0	17.6	-18.2	18.2	-19.7	18.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.2	0.5	0.7	1.8	4.8	8.5	11.4	13.3	13.2	10.5	6.4	2.7	0.6
	DBStd	5.60	4.13	3.58	3.17	2.14	2.51	2.09	1.70	1.77	1.96	2.55	3.43	3.86
	HDD10.0	1674	295	262	256	156	59	6	0	1	16	114	218	292
	HDD18.3	4421	553	495	514	405	304	208	158	160	235	371	468	550
	CDD10.0	296	0	0	0	1	14	49	101	99	31	2	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	8	0	0	0	0	0	3	3	1	0	0	0	0
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	2.9	3.3	3.1	3.3	3.0	2.7	2.3	2.2	2.2	2.8	3.2	3.5	3.6
Precipitation	PrecAvg	2661	293	218	213	163	117	96	121	171	271	341	328	329
	PrecMax	3625	544	451	401	249	293	179	225	372	489	531	587	716
	PrecMin	1912	89	78	76	16	8	35	11	43	100	149	151	151
	PrecStd	433	91	100	84	58	66	36	52	77	93	86	105	137
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.4	8.9	11.0	15.4	21.0	23.0	22.7	22.6	18.5	13.0	11.1	8.9
		MCWB	6.1	5.6	6.8	8.2	11.7	15.1	15.8	14.9	12.5	10.6	7.8	6.9
	2%	DB	7.5	7.4	8.0	12.3	17.3	19.8	20.1	19.8	16.2	12.3	9.0	7.8
		MCWB	5.8	5.2	5.6	7.3	11.0	13.7	14.9	14.7	11.7	10.2	7.7	6.3
	5%	DB	6.4	6.3	7.3	10.3	15.1	17.4	18.0	17.9	14.6	11.2	7.9	7.1
		MCWB	4.5	4.3	4.8	6.4	9.7	12.1	13.7	13.7	11.8	9.3	6.5	5.6
	10%	DB	5.8	5.6	6.2	8.8	13.0	16.0	17.1	17.1	13.7	10.0	7.3	6.0
		MCWB	4.2	4.0	3.8	5.8	9.0	11.5	13.1	13.3	11.4	8.4	5.7	4.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.6	6.8	7.8	9.2	12.7	15.7	16.4	16.3	14.4	11.7	9.0	7.7
		MCDB	8.2	7.8	9.8	13.5	17.7	22.0	21.2	20.9	15.9	12.7	9.8	8.1
	2%	WB	6.1	5.5	6.4	7.9	11.5	14.0	15.2	15.3	13.2	10.4	7.9	6.8
		MCDB	7.3	7.2	7.9	11.1	15.8	18.5	19.1	18.5	15.0	11.7	8.7	7.7
	5%	WB	5.0	4.7	5.3	6.9	10.5	12.8	14.4	14.5	12.4	9.4	7.0	5.5
		MCDB	6.6	6.5	7.2	9.5	14.0	16.4	17.6	17.4	13.9	10.7	7.9	7.0
	10%	WB	4.2	4.0	4.4	6.1	9.8	11.9	13.6	13.8	11.7	8.4	5.7	4.5
		MCDB	5.8	5.6	6.2	8.4	12.9	15.0	16.4	16.3	13.2	9.6	7.2	6.0
Mean Daily Temperature Range	5% DB	MDBR	3.8	4.8	5.5	7.1	7.6	6.9	6.2	6.4	5.8	4.8	3.9	3.8
		MCDBR	4.3	5.2	7.0	10.1	12.2	11.1	9.8	9.7	7.7	5.7	4.4	4.2
	5% WB	MCWBR	3.6	4.0	5.1	6.3	6.7	6.1	5.5	5.6	4.8	4.4	3.5	3.5
		MCWBR	4.3	4.6	6.5	8.5	9.6	9.4	8.5	8.0	5.7	4.6	4.0	3.9
Clear-Sky Solar Irradiance	taub	0.293	0.317	0.325	0.377	0.371	0.374	0.379	0.367	0.357	0.330	0.303	0.277	
		2.271	2.346	2.405	2.271	2.391	2.427	2.450	2.506	2.489	2.461	2.285	2.208	
	Ebn at Noon	586	737	837	835	867	867	854	842	795	718	563	490	
		50	71	86	114	108	106	102	90	80	63	49	40	
All-Sky Solar Radiation	RadAvg	0.44	1.21	2.38	3.91	4.91	4.91	4.23	3.56	2.32	1.27	0.57	0.32	
	RadStd	0.05	0.20	0.29	0.34	0.47	0.43	0.48	0.42	0.35	0.17	0.07	0.04	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page